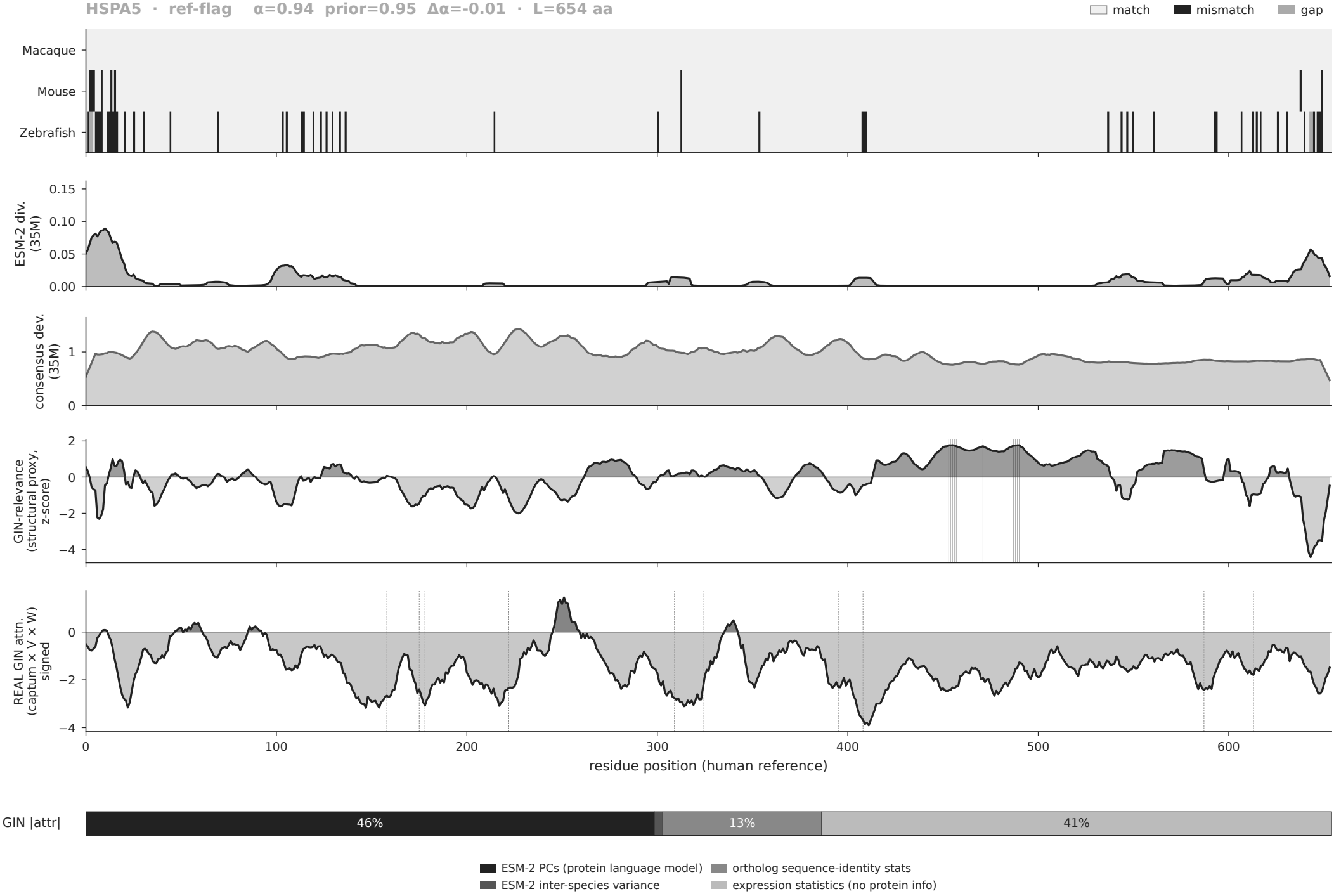

HSPA5 · ref-flag α=0.94 prior=0.95 Δα=-0.01 · L=654 aa
match
mismatch
gap
Macaque
Mouse
Zebrafish
ESM-2 div. (35M)
0.15
0.10
0.05
0.00
consensus dev. (35M)
1
0
GIN-relevance (structural proxy, z-score)
2
0
−2
−4
REAL GIN attn. (captum × V × W) signed
0
−2
−4
0
100
200
300
400
500
600
residue position (human reference)
GIN |attr|
46%
13%
41%
ESM-2 PCs (protein language model)
ESM-2 inter-species variance
ortholog sequence-identity stats
expression statistics (no protein info)